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SUSTAINABILITY REPORT

Project: Quick Analysis

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Of course. Here is the STRUCTMIND AI sustainability report for the specified structural steel package.

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Sustainability Report: 220 Tons of Structural Steel

Project Scope: 220 tons of structural steel, fabricated and delivered 900 miles via truck, finished with an SSPC 3-coat paint system. This report analyzes the environmental impact and outlines strategies for mitigation and green building certification contribution.

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1. Embodied Carbon Estimate (Cradle-to-Gate)

The production method of steel is the single largest determinant of its embodied carbon. The comparison between Electric Arc Furnace (EAF) and Basic Oxygen Furnace (BOF) is critical.

Production Method	Embodied Carbon (per ton)	Total for 220 Tons	Key Characteristic
Electric Arc Furnace (EAF)	600 - 800 kgCO■e	132,000 - 176,000 kgCO■e	Uses recycled steel scrap; powered by electricity. Strongly Recommended.
Basic Oxygen Furnace (BOF)	2,000 - 2,400 kgCO■e	440,000 - 528,000 kgCO■e	Uses virgin iron ore; energy-intensive process.

Analysis: Specifying EAF-produced steel results in a carbon savings of approximately 308,000 to 352,000 kgCO■e (308 to 352 metric tons) for this project. This is a reduction of over 70% compared to BOF steel.

• Recommendation: Mandate the use of steel from EAF mills in the project specifications.

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2. Recycled Content % Target

Recycled content is directly linked to the production method and is a key metric for circular economy principles and LEED credits.

• EAF Mills Target: >90%

• EAF steel is produced by melting scrap steel. The recycled content (post-consumer + pre-consumer) is typically between 90-98%. This is the industry standard for sustainable steel design.

• BOF Mills Target: 20-30%

• BOF steel primarily uses virgin iron ore. A smaller amount of scrap steel is added as a coolant, resulting in significantly lower recycled content.

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3. Transportation Carbon Estimate (Fabricator-to-Site)

This calculation estimates the emissions from delivering the fabricated steel to the job site.

• Distance: 900 miles

• Tonnage: 220 tons

• Emission Factor (Heavy-Duty Truck): ~165 gCO■e per ton-mile

Calculation: 900 miles × 220 tons × 165 gCO■e/ton-mile = 32,670,000 gCO■e

Total Transportation Carbon: 32,670 kgCO■e (or 32.7 metric tons of CO■e)

• Mitigation: While often fixed, sourcing from a fabricator closer to the project site is the most effective way to reduce this value.

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4. Coating VOC / Low-Emission Options

A standard SSPC 3-coat system (e.g., Zinc-Rich Primer, Epoxy Intermediate, Polyurethane Topcoat) can contain significant Volatile Organic Compounds (VOCs), which are regulated air pollutants.

• Standard System VOCs: Can range from 250 g/L to over 400 g/L.

• Low-Emission Recommendations:

• Specify Low-VOC Coatings: Mandate that all shop- and field-applied coatings comply with stringent environmental standards, such as SCAQMD Rule 1113 (<100 g/L for most industrial coatings).

• High-Solids Formulations: Specify high-solids epoxy and polyurethane coatings, which have less solvent and lower VOC content per unit of coverage.

• Powder Coating: For smaller components or sections where feasible, powder coating is a zero-VOC alternative that offers excellent durability.

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5. Material Efficiency & Waste Factor

Efficient fabrication is key to minimizing waste and the associated environmental impact.

• Typical Waste Factor: 3-5% for structural steel fabrication. For 220 tons, this equates to 6.6 - 11 tons of scrap.

• Optimization Strategies:

• BIM Integration: Utilize the Building Information Model (BIM) for direct-to-fabrication outputs, minimizing errors and optimizing material cuts.

• Nesting Software: Require the fabricator to use advanced nesting software to lay out cuts on raw steel sections, maximizing yield and minimizing off-cuts.

• Design for Standard Lengths: Where possible, design members to utilize standard mill-produced lengths to reduce cutoff waste.

• Circularity: A major advantage of steel is that 100% of the fabrication scrap is valuable and can be collected and recycled back into the EAF steelmaking process.

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6. EPD / LEED / BREEAM Contribution Summary

By specifying EAF steel and requiring proper documentation, this project can make a significant contribution to green building certifications like LEED v4.1.

Action Item: Require a product-specific, Type III Environmental Product Declaration (EPD) from the steel mill/supplier.

LEED v4.1 MR Credit Contribution:

• MRc: Building Life-Cycle Impact Reduction (Option 4, Whole-Building LCA):

• MAJOR CONTRIBUTION. Using EAF steel with its low embodied carbon value will dramatically reduce the project's Global Warming Potential (GWP) in a whole-building LCA, making it much easier to achieve the required 10% reduction compared to a baseline (which would assume industry-average or BOF steel).

• MRc: Building Product Disclosure and Optimization - EPDs (Option 1):

• CONTRIBUTES 1 PRODUCT. Providing a product-specific EPD for the 220 tons of structural steel contributes one product towards the 20 needed for this credit.

• MRc: Building Product Disclosure and Optimization - Sourcing of Raw Materials (Option 2):

• STRONG CONTRIBUTION. EAF steel with >90% recycled content (mostly post-consumer) will significantly contribute to this credit. The value of the steel package is multiplied by its recycled content percentage, helping the project meet the threshold of 25% (by cost) of total building materials.

• EQc: Low-Emitting Materials:

• Specifying and documenting the use of low-VOC coatings for all shop and field applications will contribute to earning this credit.

Summary: The strategic specification of EAF-produced structural steel, supported by an EPD and paired with low-VOC coatings, is one of the most impactful decisions for improving the sustainability profile of the building structure and maximizing contributions to LEED Materials & Resources (MR) credits.